Erdem Aslan

AI Solutions Engineer | Applied AI: integrations, agents and evals

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AI Solutions Engineer at Tam Hangers: I work with department teams to find repetitive work AI can take over, then build and integrate the tools for it. In my own projects I practice forward-deployed engineering against simulated customer systems: agents with guardrails enforced in code, retry-safe integrations and evals that gate CI. Turkish native, English C1.

WORK EXPERIENCE

AI Solutions Engineer | Tam Hangers, Istanbul (on-site) Aug 2026 – Present

Joined as an IT Specialist intern (Jul – Aug 2026), then moved straight into this full-time role.

- Map department workflows with the teams and build AI automations and internal tools for their repetitive, manual steps.

Full-Stack Developer Intern | Proceedit Trading, Barcelona, Spain (remote) Oct 2025 – May 2026

- Built Python/Flask and PostgreSQL backend components: REST endpoints and database integrations.
- Supported AWS/Kubernetes operations and production analysis of CatBoost/LSTM models on financial time series.
- Worked in an international Agile team (JIRA); prototyped mobile screens in FlutterFlow and Figma.

iOS Developer Intern | Lideatech Jun 2025 – Jul 2025

- Built the FinFlow personal-finance app with SwiftUI, Firebase (Firestore, Authentication), MVVM and interactive dashboards.

Game Operator & Cook | Morey's Piers, USA (Work & Travel) Jun 2024 – Sep 2024

SELECTED PROJECTS

VoiceOps FDE Sandbox | FastAPI · tool calling · RAG · PostgreSQL · Docker 2026

Simulated customer deployment: a Turkish/English voice and chat agent for a fictional bank's contact center, calling its mock CRM, core banking, helpdesk and scheduling systems over REST. Identity checks, confirmations and supervisor approvals are enforced in code. 52 of 55 golden scenarios pass in CI (mock LLM and speech, synthetic data).

Integration Reliability Lab | FastAPI · webhooks · PostgreSQL · Redis 2026

Ordering hub for a fictional café chain that keeps mock POS, payments and loyalty systems in sync when webhooks arrive twice, late, out of order or forged: signed webhooks, idempotency keys, an outbox with retries, dead letters, reconciliation and an operator runbook. Six injected failure scenarios pass in CI; locally, 0 of 50 double credits after retried timeouts.

Agent Eval & Data Health Hub | DuckDB · FastAPI · Streamlit · CI gate 2026

Evals, data-quality checks and a CI regression gate for AI agents. Against the VoiceOps API (mock LLM) it found four problems VoiceOps' own evals had missed, one of them reading another customer's transactions by ID alone; all four were fixed and rechecked.

LLM Distillation & Domain Specialization | Bachelor's thesis · arXiv:2601.16219 (preprint) 2025

Distillation into Qwen-2.5-7B (Unsloth, 4-bit QLoRA) with three data strategies: the 15,000-line general and 2,000-line unstructured sets kept hallucinating; a 500-line context-grounded synthetic set scored 96.7% on the 30-question general test (90% on regulation questions); training fit in 16 GB of GPU memory instead of 40+ GB.

Historai | FastAPI · Next.js · Groq · Gemini · SSE 2026

Alternate-history simulator on live LLM APIs: 6–8 LLM-played historical actors decide each turn in parallel (Groq, Gemini as fallback); progress streams to a Next.js UI over SSE, and each run ends in a narrative report and a timeline.

EDUCATION

B.Sc. Computer Engineering | Düzce University Jun 2026

GPA 3.17 / 4.00 · Completed the four-year curriculum in 3.5 years · One-year English preparatory program

Erasmus+ exchange, Computer Science | Gdańsk University of Technology Feb 2026 – Jun 2026

SKILLS, LANGUAGES AND TRAINING

LLM applications	tool calling and agents, RAG, guardrails and approval gates in code, prompt versioning, golden-set evals, CI regression gates, Groq and Gemini APIs
Integrations / backend	Python, FastAPI, Flask, REST, signed webhooks, idempotency, outbox, retries and dead letters, reconciliation, PostgreSQL, Redis, DuckDB
DevOps / cloud	Docker, GitHub Actions, pytest, AWS, Kubernetes, OpenTelemetry, Linux
Model work	knowledge distillation, LoRA/QLoRA, Unsloth, PyTorch, Hugging Face, transfer learning (EfficientNet)
Also	TypeScript, Next.js, Swift/SwiftUI, Firebase
Languages	English: full professional proficiency (C1) · Turkish: native
Training	Mustafa Akgül Open Source Software Camp: Application Monitoring with OpenTelemetry (Prometheus, Grafana, Kafka), Feb 2026